

Darsh Shah

Senior Software Development Engineer | Java/Python/Typescript | AWS CDK

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Summary

Software Development Engineer 2 at Amazon with 4+ years of experience building large-scale cloud infrastructure and AI-driven modernization platforms. Proven technical leader with end-to-end ownership of distributed systems, mentoring junior engineers, and driving cross-team initiatives. Specialized in AWS, Java, and Python, with deep expertise in high-concurrency architectures, CI/CD automation, and production reliability at scale.

SKILLSET

- Languages: Java, Python, TypeScript, Scala, SQL
- AI & Modernization: LLM-driven Systems, Agentic AI Architectures, Claude (Anthropic), Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Orchestration, Code Refactoring Automation
- Backend & Systems: Microservices, Distributed Systems Design, High-Concurrency Architectures, REST APIs, System Design, State Management, Control Planes
- Cloud & Data: AWS (EC2, ECS, ECR, VPC, IAM, Cloudwatch), Docker, DynamoDB, SonarQube, Kubernetes (EKS)
- Leadership & Execution: Technical Mentorship, End-to-End Ownership, Cross-Team Collaboration, Stakeholder Communication, Roadmapping & Technical Planning

WORK EXPERIENCE

Senior Software Development Engineer, Amazon Web Services(Agentic AI) Jul 2024 - Present

- Architected **AWS Transform's multi-agent orchestration platform**, enabling autonomous modernization of legacy enterprise systems to cloud-native Java architectures, supporting **23+ enterprise engagements and ~10M+ lines of production code refactoring**.
- **Led development of ATX Reforge**, a production-grade LLM-powered refactoring engine, designing distributed task agents and control-plane orchestration that processed **millions of lines of code per modernization cycle** with deterministic reliability.
- **Designed high-concurrency Agent-to-Agent (A2A) protocols and stateful control plane services**, ensuring session integrity and consistent execution across parallel AI workloads under heavy load.
- **Engineered an intelligent LLM capacity management and rate-limiting layer**, implementing exponential backoff and adaptive throughput control to maintain **100% service availability during model capacity constraints**.
- **Built high-throughput FastAPI-based control and health services**, enabling real-time monitoring of task agents and reducing failure detection time by **60% through automated health telemetry and alerting**.
- Led cross-functional collaboration across 3+ engineering teams, driving architectural alignment on agent orchestration patterns and establishing shared standards that accelerated delivery timelines by 30%.
- Influenced system architecture and operational strategy across teams, designing high-throughput **REST APIs** for orchestration and health monitoring of distributed AI services, improving reliability and reducing operational overhead by **40%**.

Software Development Engineer, Amazon Web Services(Agentic AI) Jan 2022 - Jun 2024

- **Built the Regression Testing Snapshot (RTS) Manager**, automating Oracle, PostgreSQL, and Sybase database restoration workflows, reducing environment recovery time from **4 hours to under 3 minutes (98% improvement)** across enterprise testing pipelines.
- **Developed scalable Java and Python backend services** supporting AWS Transform modernization workflows, handling large enterprise applications with strict latency and reliability requirements.
- **Designed and provisioned production-grade AWS environments using AWS CDK**, implementing secure VPCs, IAM, KMS encryption, and observability tooling across multi-account deployments - [AWS SDE Documentation](#).
- Designed event-driven infrastructure lifecycle automation (**Lambda + EventBridge**) and integrated CI/CD observability with Slack and CloudWatch canaries, reducing infrastructure costs by **50%** and improving production issue detection speed by **40%**.
- Improved database performance by refactoring **SQL queries and indexes**, reducing query execution time by 30%.

Early Experience: Software Engineering Intern – AT&T Labs | Copart

May 2021 - Dec 2021

EDUCATION

Masters of Science: Computer Science - UT Dallas(GPA 3.82/4.0)

Jan 2020 - May 2022

Bachelors of Engineering: Computer Engineering - GTU(GPA 8.46/10)

Aug 2015 - May 2019

Certifications

AWS Solutions Architect Associate

Dec 2024